Compute-Efficient RL Post-training for Low-Resource MT

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Abstract

Group-relative RL such as GRPO is widely used for LLM post-training, but its compute efficiency for low-resource machine translation is unclear. We study whether this compute can be reallocated across tokens and rollouts without sacrificing quality. We propose selective token updates, which target update cost, and reasoning-prefix reuse, which targets rollout cost, and evaluate both on low-resource English→xx directions from WMT26 against equal-budget SFT. We deliver a compute–quality frontier, a QE-reward audit, and reproducible code and compute logs.

1. Introduction

Group-relative RL methods such as GRPO (Shao et al., 2024) and DAPO (Yu et al., 2025) are widely used for LLM post-training, but their compute efficiency for low-resource machine translation (MT) is unclear. Both stages of RL are expensive: the model generates reasoning tokens and final translations for every rollout in a group, a verifier scores each one, and gradients flow through long sequences. Under an outcome-only reward, much of this computation may produce noisy gradients.

We ask: *can we reduce the compute of group-relative RL for low-resource MT by reallocating it across tokens and rollouts, without sacrificing translation quality?* Keeping GRPO critic-free with a fixed QE reward, we contribute **selective token updates** (targeting update cost) and **reasoning-prefix reuse** (targeting rollout cost). Our hypothesis is that outcome-only RL miscredits many reasoning tokens, while final-translation tokens and a subset of reasoning tokens carry most of the useful signal. The interventions build on high-entropy token selection (Wang et al., 2025) and prefix-conditioned RL (Setlur et al., 2026).

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2. Compute Budget

We split the cost of a training run into two budgets: B_{test}, forward compute for rollouts and reward scoring, covering reasoning tokens, final translations, and MetricX calls; and B_{back}, compute for parameter updates, i.e. forward and backward passes over the tokens that receive gradient. We count both in FLOPs, using $2N$ per token for forward-only passes and $6N$ per token for gradient steps of a model with N parameters (MetricX counted with its own size), and cross-check the estimates against measured GPU hours. For algorithm $\mathcal{A} \in \{\text{SFT}, \text{GRPO}, \text{DAPO}\}$ and reasoning intensity κ (the reasoning-token budget per rollout: none, short, or long), a configuration costs $B_{\mathcal{A},\kappa} = B_{\text{test}} + B_{\text{back}}$ and reaches evaluation performance $P_{\mathcal{A},\kappa}$. Against an equal-budget SFT baseline $\mathcal{R}$ ($B_{\mathcal{R}} = B_{\mathcal{A},\kappa}$) and a lower-budget anchor $\mathcal{R}_0$ (the shared SFT warm-up), we report

$$\Delta P = P_{\mathcal{A},\kappa} - P_{\mathcal{R}}, \qquad \eta = \frac{P_{\mathcal{A},\kappa} - P_{\mathcal{R}_0}}{B_{\mathcal{A},\kappa} - B_{\mathcal{R}_0}}.$$

ΔP tests whether RL beats SFT at equal total compute; η measures quality gained per additional unit of compute.

To build $\mathcal{R}$, SFT receives the same total FLOPs as the RL run it is compared with, spent on more parallel data or more epochs; if SFT saturates earlier, we report its best checkpoint and the unused budget. Rather than single end points, we record P against cumulative B throughout training, so ΔP and η can be read at any budget and methods are compared along their whole compute–quality curves. The two interventions act on different terms: token selection targets B_{back} and prefix reuse targets B_{test}. We log both terms per run, together with GPU hours and peak memory.

3. Methods

Selective token updates. Standard GRPO applies the group-relative advantage A_i to every generated token. For rollout i, token t, and importance ratio $\rho_{i,t}$, a binary mask $M_{i,t}$ gives

$$\mathcal{L}_{\text{mask}} = -\frac{\sum_{i,t} M_{i,t} \min\big(\rho_{i,t}A_i, \text{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon)A_i\big)}{\sum_{i,t} M_{i,t}}.$$

We compare (i) full-token GRPO; (ii) translation-only updates; and (iii) all translation tokens plus reasoning tokens

kept independently with retention rate q. Masks are sampled independently of rewards, so the masked loss is a random subsample of the full objective and introduces no reward-dependent selection bias into the group comparison. A loss mask alone changes the learning signal but barely reduces backward FLOPs, since gradients still flow through masked positions via attention. To realize B_{back} savings, regime (ii) encodes the reasoning prefix without gradient and backpropagates only through translation tokens; for regime (iii) we report gains in η separately from measured per-step savings.

Reasoning-prefix reuse. Reasoning prefixes dominate rollout length. For source x, we generate or retrieve $K < G$ prefixes $\{p_1, \ldots, p_K\}$; rollout i selects p_{k_i} and samples only the translation, $y_i \sim \pi_\theta(\cdot \mid x, p_{k_i})$. Rewards score only y_i, and advantages are computed within the group as usual. Prefix cost scales with K instead of G, or vanishes when prefixes come from a replay buffer; construction and storage overhead still count toward B_{test}. Since shared prefixes can collapse group diversity, stored prefix tokens are randomly kept, dropped, or replaced by mask tokens before use; a separate variant applies this masking to the final response. Prefix reuse composes naturally with regime (ii): a shared prefix is encoded once, without gradient, for all rollouts using it.

4. Data, Models, Reward, and Evaluation

Data and models. We target low-resource English→xx directions covered by WMT26 (Kocmi et al., 2026), training on WMT26 general-MT data plus public parallel data collected with MTData (Gowda et al., 2021). We use Qwen3.5-0.8B and 2B, scaling to 4B if compute permits (Qwen Team, 2026); larger Qwen3.5 models (27B dense, 35B-A3B MoE) serve as untuned reference systems. A pilot on WMT26 development data measures (i) the small models' headroom per direction, (ii) intra-group reward variance at $G = 8, T = 1.0$ (zero-variance groups give GRPO no signal), and (iii) the gate failure rate. These fix the language set, G, K, and q.

Fixed outcome reward. For source x and translation y,

$$r(x, y) = \mathbf{1}_{\text{gate}}(x, y) \left(1 - \frac{\text{MetricX-24-QE}(x, y)}{25}\right),$$

using MetricX-24-Hybrid-Large in reference-free QE mode (Juraska et al., 2024). The gate checks non-empty output, fastText target-language ID, and length-ratio bounds, filtering degenerate rollouts like source copying. The reward is fixed, not the contribution; no learned critic or reward model is used. We audit its rankings against WMT26 cESA annotations (Zouhar et al., 2026b) and hallucination sensitivity. We also compute a reference-based diagnostic $V_{\text{ref}}(x, y, y^*) = \text{MetricX-24}(x, y, y^*)$, but never train on it. If efficiency gains disappear under V_{ref}, they depend on the QE reward; if they persist, they are robust to the choice of reward.

Evaluation. As training optimizes a QE metric, headline results must not rest on QE alone. Held-out WMT26 test sets are scored with COMETKiwi22 (Rei et al., 2022b), together with reference-based chrF++ (Popović, 2017) and COMET-22 (Rei et al., 2022a). Final systems are compared with cESA human evaluation (Zouhar et al., 2026b) on a sample of test segments, which ranks several systems' translations of the same source side by side. LTB (Zouhar et al., 2026a) diagnoses rule-based verification failures. FLORES+ is excluded from headline results due to contamination concerns.

5. Experiments

I. Compute–quality frontier: SFT vs. GRPO vs. DAPO under matched $B_{\mathcal{A},\kappa}$, reporting ΔP and η; this establishes whether RL is worth its cost before any intervention. **II. Selective updates:** the three mask regimes across q, measuring B_{back}, quality, and training stability. **III. Prefix reuse:** no reuse, full reuse, and stochastic-masked reuse across K/G, measuring B_{test} and quality. **IV. Combination and ablations:** both interventions jointly vs. each alone, with a small sweep over κ, and a failure–gain matrix over source copying, off-target language, omissions, and named-entity errors.

All runs share initialization, data, evaluation decoding, and reward, except for the factor being varied. Hyperparameters are tuned only on development data, with multiple seeds where feasible; rollouts and compute logs are archived.

6. Risks and Fallbacks

RL does not beat SFT at equal compute. The frontier from Experiment I is then itself the main result, and the interventions are still compared within RL by η. **Too little group variance.** If small models produce many groups with identical rewards, we use DAPO's dynamic sampling and raise T or G, as guided by the pilot. **Prefix reuse collapses diversity.** We track intra-group reward variance as a function of K/G and the masking rate, and report the point where reuse stops paying off. **Reward hacking.** The gate, reference-based metrics, and cESA expose gains that do not transfer; we report them as findings rather than changing the reward mid-study.

7. Expected Outcomes

A compute–quality frontier for SFT, GRPO, and DAPO on low-resource English→xx MT; evidence for or against token selection to cut B_{back} and prefix reuse to cut B_{test}; a reward audit separating genuine efficiency gains from QE-reward artifacts; and a failure–gain matrix showing which errors RL fixes. Deliverables are reproducible code, configs, checkpoints, evaluation scripts, compute logs, and a paper-style report. A well-supported negative result is acceptable.

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